

Claude Code Prompt

World Bank Open Data — LibreOffice Calc Add-In

Objective

Build a free, open-source LibreOffice Calc add-in (UNO Add-In component) that retrieves indicator data from the World Bank Open Data Indicators API and exposes it through spreadsheet cell functions. The extension packages as a single `.oxt` file and targets cross-platform use (Slackware, Debian, Ubuntu, Windows).

Hard Constraints (LibreOffice-embedded JVM)

These are non-negotiable and must be respected throughout:

- **Java 8 source/target only.** No records, no var, no `java.net.http.HttpClient`, no switch expressions, no text blocks.
- **Zero third-party dependencies.** Everything from the JDK 8 standard library only. Use `HttpURLConnection` for networking and a hand-rolled minimal JSON parser (do NOT bundle Jackson/Gson — bundled jars risk classloader conflicts inside LibreOffice's JVM).
- **Add-In only pattern.** Implement as a Calc Add-In exposing cell functions; no sidebar, no dialog, no Calc document manipulation beyond function return values.
- **Non-blocking cell functions.** Cell functions must NEVER block on network I/O. Use a cache + background-fetch pattern: on a cache miss, kick off an async fetch, return a sentinel immediately, and let the user recalculate (`Ctrl+Shift+F9`) to pick up the resolved value.
- **CalcAddIn CompatibilityName.** Every exposed function MUST have a `CompatibilityName` set in `CalcAddIn.xcu`. Omitting this causes silent data loss when the sheet is saved as XLS/XLSX and reopened. This is a recurring critical detail — do not skip it.

The World Bank Indicators API

No API key required. REST endpoints returning JSON when `?format=json` is appended (default is XML).

Base URL:

`https://api.worldbank.org/v2/`

Core endpoint — indicator values for one or more countries:

`https://api.worldbank.org/v2/country/{codes}/indicator/{id}?format=json`

Example — US GDP (current US\$), indicator NY.GDP.MKTP.CD:

`https://api.worldbank.org/v2/country/US/indicator/NY.GDP.MKTP.CD
?format=json&per_page=20000`

API quirks that must be handled explicitly

- **Two-element array response.** The JSON top level is an array: element `[0]` is a pagination header object (page, pages, per_page, total); element `[1]` is the array of observation objects. The parser must index into `[1]` for data and must handle the case where `[1]` is null (no data / bad indicator).
- **Observation shape.** Each observation has `indicator{id, value}`, `country{id, value}`, `countryiso3code`, `date` (year as string), `value` (number or null).

- **Nulls are common.** `value` is frequently null for years with no data — must map to empty/NA, not zero.
- **Pagination.** Default `per_page` is 50. Use `per_page=20000` to get a full series in one request, or read pages from the header and loop.
- **Multiple countries:** join ISO2 (or ISO3) codes with a semicolon, e.g. `country/US;GB;DE`. `country/all` returns every country.
- **Date filtering:** `&date=2000:2023` for a range, or `&date=2020` for one year.
- **MRV (most recent values):** `&mrv=5` returns the latest 5 non-null observations — ideal backend for a 'latest value' function.
- **Errors.** An invalid indicator/country returns a JSON array whose [0] element contains a message array rather than pagination fields. Detect and surface as an error sentinel.

Cell Functions to Expose

Each function needs a `CompatibilityName` in `CalcAddIn.xcu`. Suggested set:

- **WBVALUE(country, indicator, [year])** — single scalar. If year omitted, return the most recent non-null value (uses `mrv=1`). Returns a number or NA.
- **WBSERIES(country, indicator, [startYear], [endYear])** — spilling array of [year, value] pairs (two columns) for the range. Sorted ascending by year.
- **WBLATEST(country, indicator)** — convenience wrapper returning the most recent value plus its year (two cells: value, year), useful for dashboards.
- **WBMETA(indicator)** — returns the indicator's human-readable name/description (from the `/indicators/{id}` metadata endpoint), for labeling.

Consider whether array-returning functions should return a proper sequence (`Object[][]`) so they spill across cells; document the expected selection/CSE behavior for the Java 8 / older-Calc target.

Caching & Background Fetch

- In-memory cache keyed by the normalized request URL (country+indicator+date+mrv).
- On cache hit with fresh data: return immediately.
- On cache miss: enqueue a background fetch on a small bounded thread pool, store a PENDING marker, and return a sentinel (e.g. the string `#FETCHING` or an NA) so the cell never blocks.
- When the background fetch completes, replace PENDING with the parsed result. The user recalculates to pick it up.
- Cache entries should carry a timestamp; optionally expire after a configurable TTL (World Bank annual data changes rarely, so a long default TTL is fine).
- Thread pool must be daemon-threaded and shut down cleanly on component disposal.

Project Structure & Deliverables

Produce a complete, buildable project:

- `src/` — Java 8 sources: the UNO Add-In implementation class, the minimal JSON parser, the HTTP client wrapper, the cache/fetch manager, and the XWorldBank IDL-backed interface.
- `idl/` — the `.idl` defining the add-in interface, plus notes on compiling with `idlc` and `regmerge`.

- META-INF/manifest.xml, description.xml, CalcAddIn.xcu (with all CompatibilityNames), and the type registry (.rdb).
- A build script (plain shell + javac/jar, no Maven inside the extension) that produces the .oxt, plus clear step-by-step build instructions listing the LibreOffice SDK tools required (idlc, regmerge, unopkg).
- README.md — installation via unopkg add, usage examples for each function, the list of common indicator codes, and the recalculation note for pending fetches.
- LICENSE — MIT.

Additional Notes

- Keep the JSON parser deliberately small — it only needs to handle the World Bank response shape (arrays, objects, strings, numbers, null, booleans). Do not attempt a general-purpose parser beyond what these responses require, but do handle nested objects and escaped strings correctly.
- Set a sane connect/read timeout on HttpURLConnection and a descriptive User-Agent header.
- Gracefully degrade on network failure: return an error sentinel, never throw out of a cell function.
- Before writing any code, restate the locked architectural decisions and confirm the function signatures. Then produce the full implementation.

License: MIT • Target: LibreOffice Calc, Java 8 embedded JVM • Dependency-free